Imperatives of Small & Micro Nuclear Reactors for a Developed & De-carboned India

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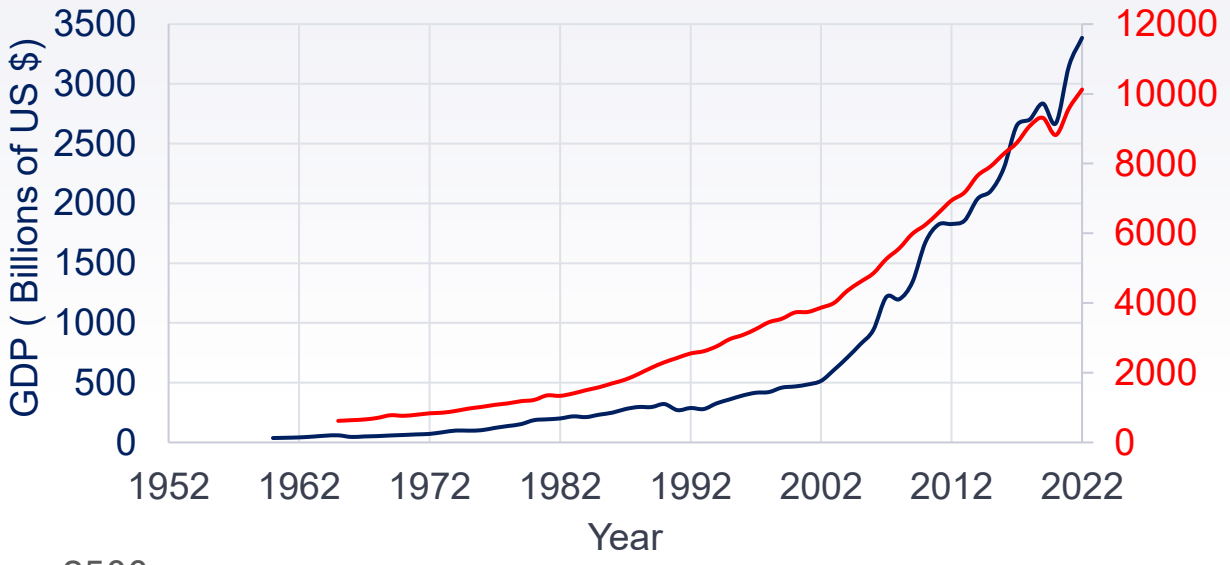
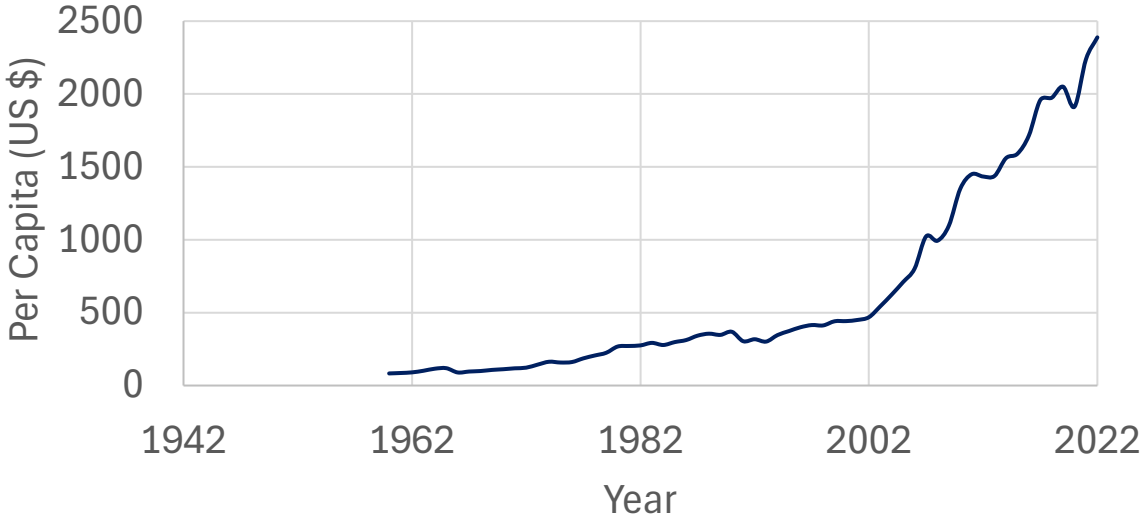
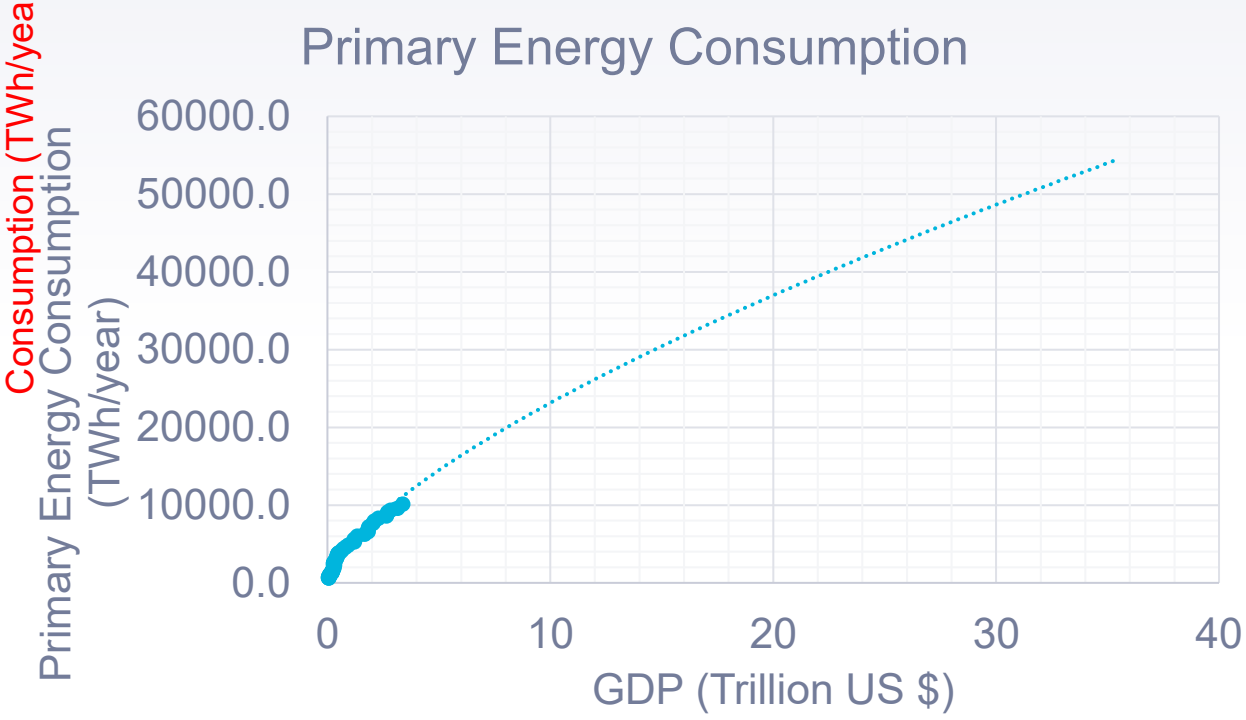
What we understand by a Developed India?

1. Current India's per capita income is close to $2,500 and must surpass $21,664 (8 times) (according to World Bank) for the nation to be classified as a high-income nation.
2. Today, India’s economy is close to $ 4 trillion and to grow by $ 35 trillion @2047 to reach the status of developed India
3. For this India must grow by 7.6% per year for continuous 25 years (CAGR of 10.6%, which incorporates the impact of inflation) (According to a study by the Economic Research Department of the Reserve Bank of India).
4. To accomplish the GDP growth rate, a sustained energy growth to support the GDP is a must, while considering the climate change and India’s commitments to reach net zero at 2070 by a de-carboned energy sector.

Linkage Between GDP Growth Rate vs Energy Consumption





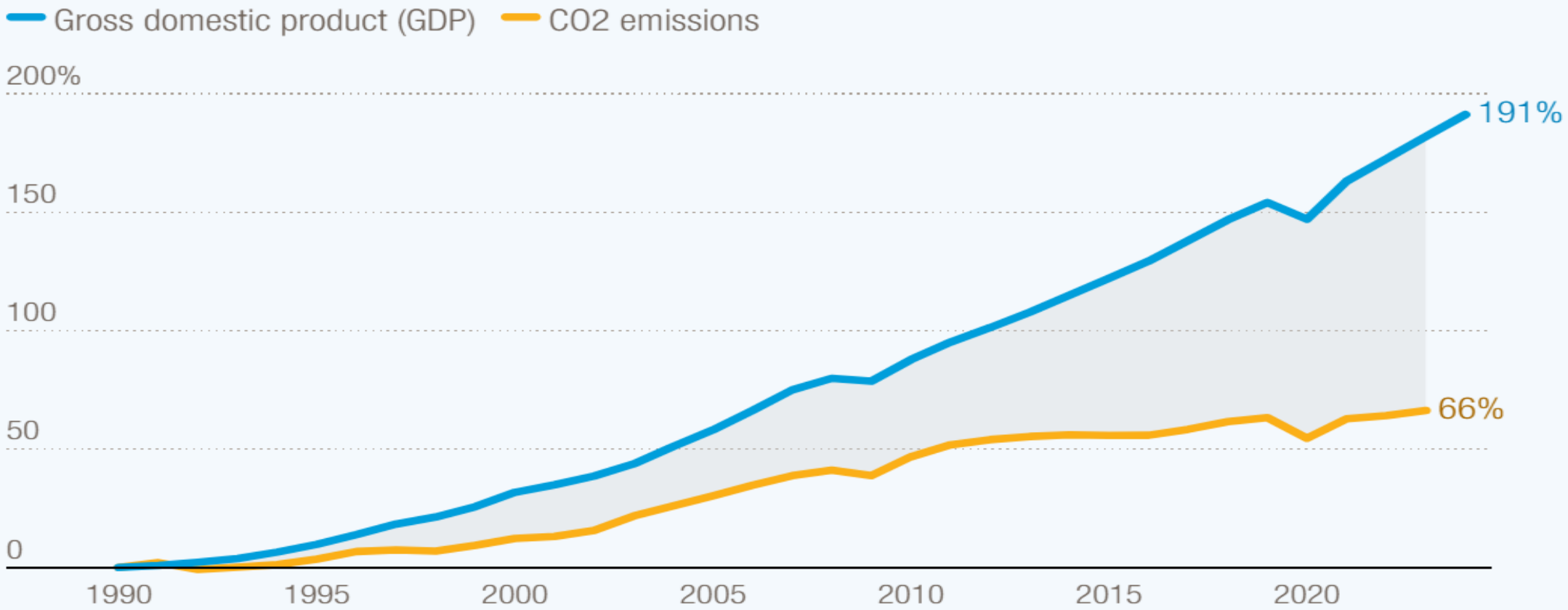


To support a $35 trillion economy, the envisaged energy requirements can be at least of the order of 50000 to 60000 Terawatt hour per year.

https://www.macrotrends.net/countries/IND/india/gdp-gross-domestic-product

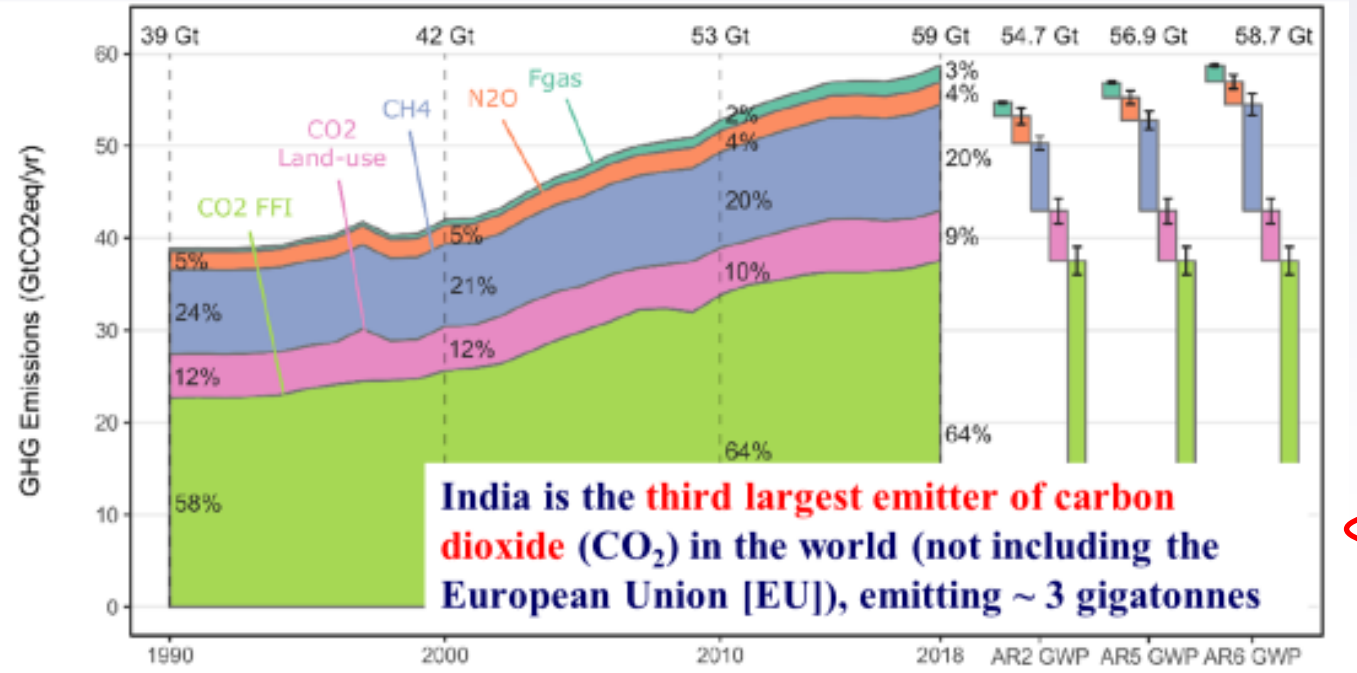
Historic GDP Growth vs. CO2 Emissions

Change in CO2 emissions (percentage) and gross domestic product (GDP), 1990–2024



Source: UN Trade and Development (UNCTAD) data based on World Bank, 2025 and Global Carbon Budget, 2024
Note: GDP is measured in constant 2021 international dollars (PPP). World CO2 emissions is estimated based on annual consumption-based CO2 emissions
Get the data
Download image

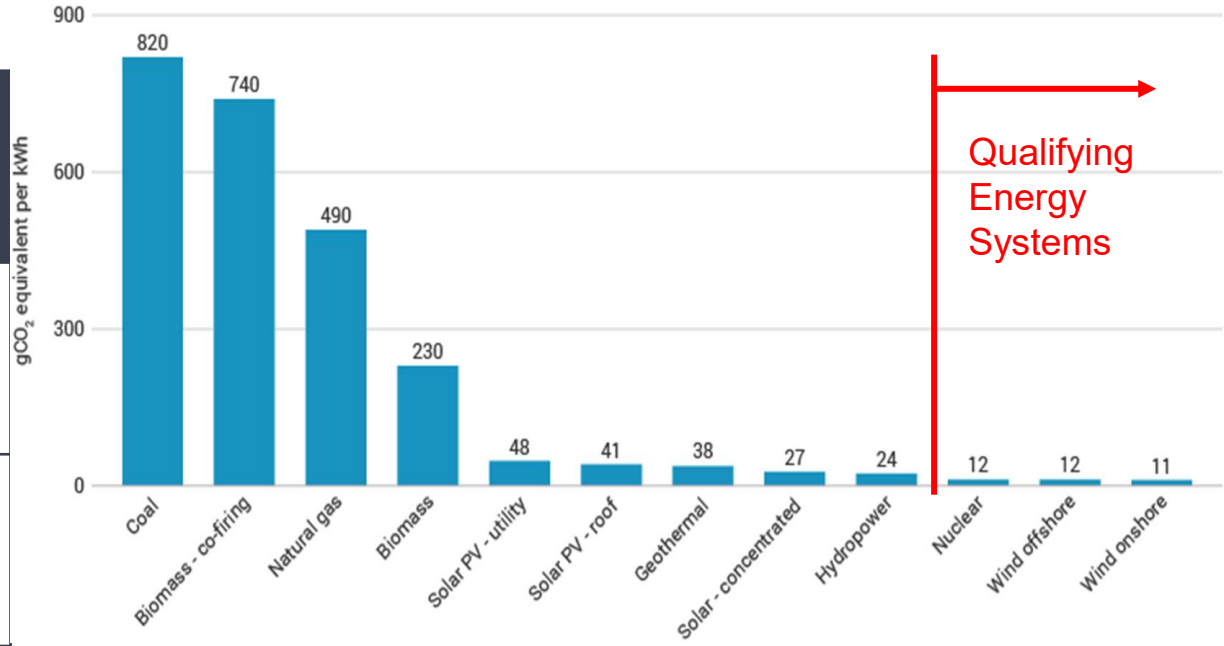
Carbon Emissions – To be Reduced and Fast



CO_2 Emissions and Emission Targets

Country	2017 CO_2 Emissions from Electricity (gCO_2/kWh)	Percentage of Nuclear in the Energy Mix	2050 IEA Energy Technology Perspectives 2°C Scenario (gCO_2/kWh)	MIT study for Deep De-carbonisation (gCO_2/kWh)
United States	470	18 %	11	1
China	680	2.2 %	24	1
United Kingdom	350	15 %	11	1
France	90	68 %	11	1
India	815	3 %	-	-

Global Temperature Limit	CO_2 Budget (Giga Tons)	Time Left
1.5 °C (50 to 67% Probability)	400 to 500	8 – 10 Years
2 °C (50 to 67% Probability)	1150 to 1350	~ 20 Years



Can a Developed India get Decarboned by Renewables Alone?

What is the maximum potential off renewables in India?

Renewable source	Prerequisite	Total generation TWh/annum
Solar	Utilization of 10% of barren and uncultivated land in India (30000 Sq km), (Sukhatme et al), Storage technology available, smart grids in place.	2000-3000
	MNRE - solar potential of the country is around ~ 1640 TWh/annum assuming 3% of the waste land (~ 3% of 500000 Sq km = 15000 Sq km) area to be covered by Solar PV modules.	
	NITI Aayog - potential to be around 2040 TWh/annum with storage technology available and smart grids in place.	
Wind	100 meter mast height, smart grids and storage in place (MNRE)	978
Hydro	All the 150 GWe potential utilized with 60 % PLF (CEA)	788
Bio	All the potential utilized (MNRE)	60
	Total	**3826 - 4826**

How Much N-Power India would Require?

1. The total potential for India from renewables is hardly close to 4000 Terawatt hour per year, which is just 40 % of India's today's total energy consumption.
2. Assuming the developments in the energy efficiency, smart grids and excellent storage systems in place, the capacity may rise to a maximum of ~8000 Terawatt hour per year.
3. Even for today's total energy De-carbon requirement; the Balance from Clean Energy will be ~ 2000 Terawatt hour per year.
4. And for India's further development to reach a $35 trillion economy by 2047, the balance from clean energy requirement will be of the order of several thousand Terawatt hour per year (more than 2500 GWe) apart from utilisation of full potential of renewables.

Coal to Nuclear Transition – The American Strategy

COAL *to* NUCLEAR

Repowering coal plants with advanced nuclear reactors can help unlock **new job, economic, and environmental opportunities** for energy communities across the country as the United States shifts toward cleaner energy sources. Here's how it works.

Retire | *Reuse* | *Re-Power*

30% of the nation's coal plants are expected to retire by 2035

more than 300 existing and retired coal plants are suitable to host advanced nuclear plants and technologies

up to 35% can be saved on plant construction costs by reusing existing coal plant infrastructure

650+ high-paying nuclear jobs would be created or converted in the region

up to 86% drop in emissions in the surrounding region by replacing coal with nuclear plants

$275m added annual economic activity in nuclear-sector communities

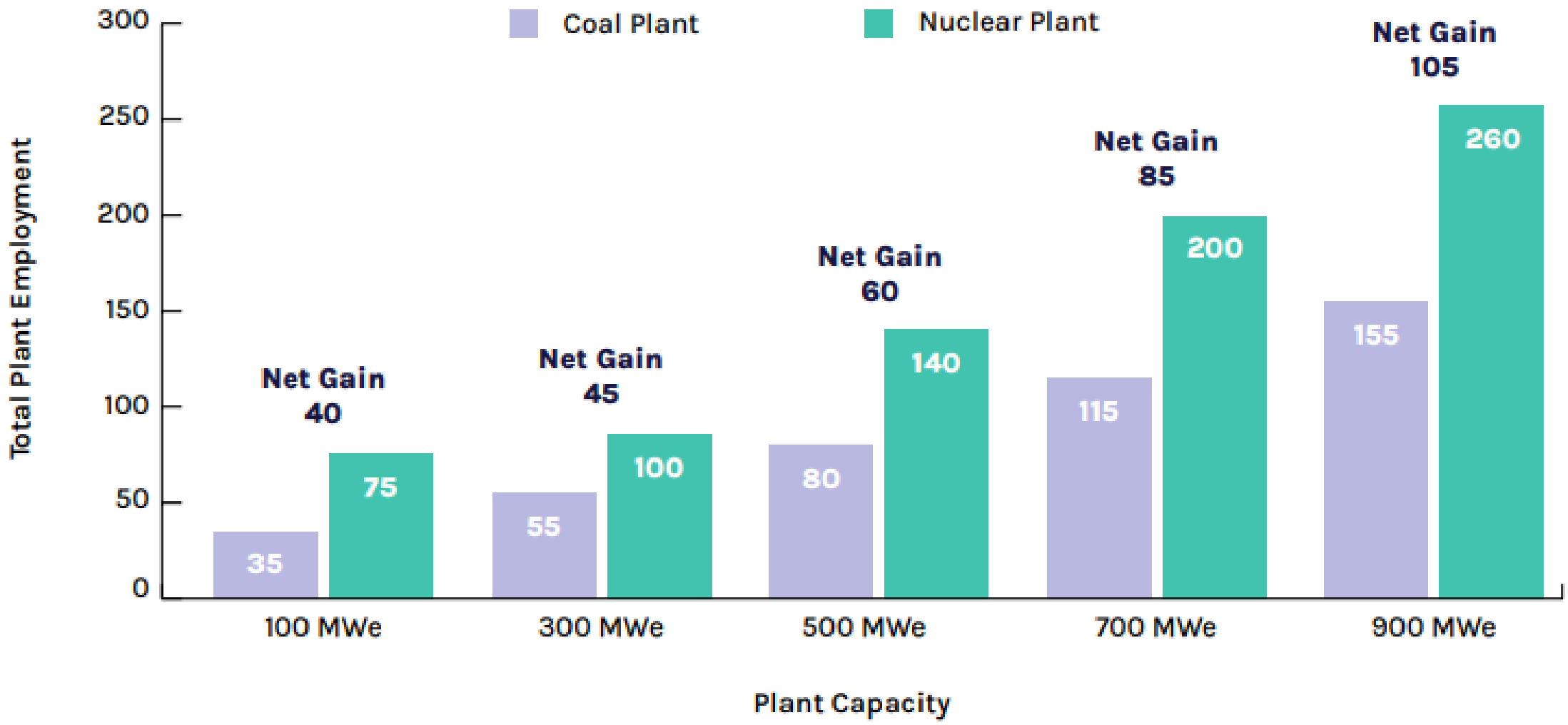
POWER & OPERATIONS

Nearly 400 coal sites could be home to the next 250 GW of U.S. nuclear capacity

Thu, Sep 15, 2022, 8:00PM | Nuclear News

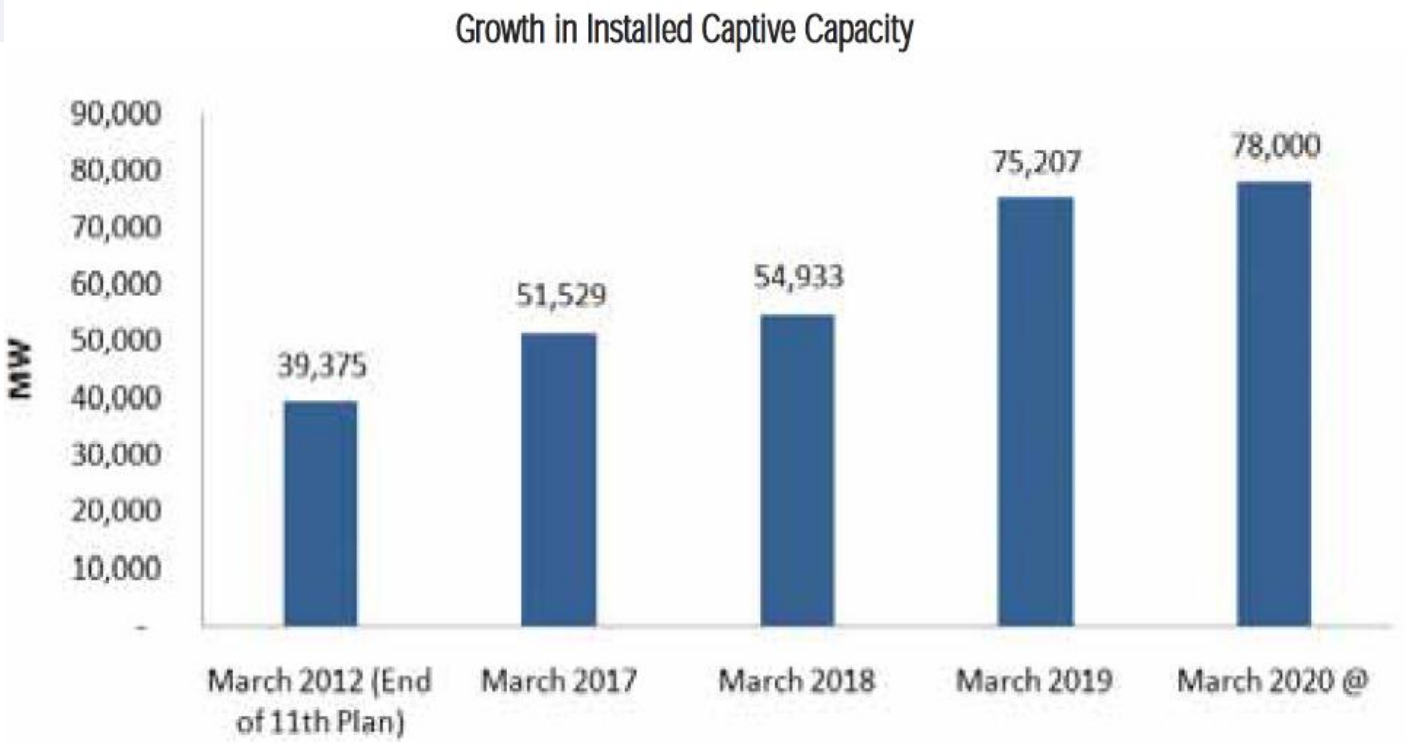
The Naughton coal-fired power plant near Kemmerer, Wyo., has two units set to retire in 2025 and be replaced by a TerraPower Natrium reactor. (Photo: PacifiCorp)

ESTIMATED EMPLOYMENT BY GENERATION



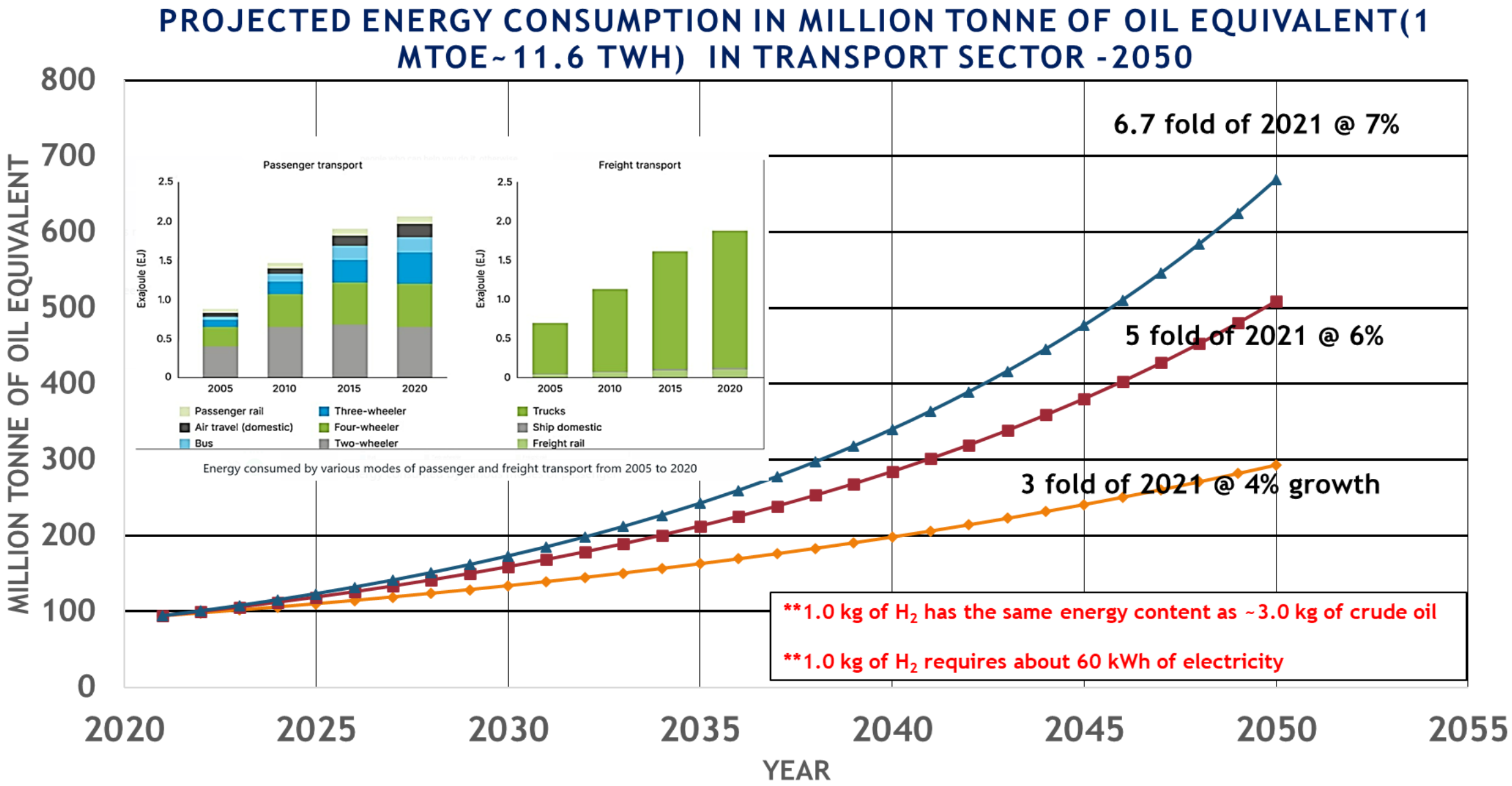
SMRs Demand for Captive Power Plants for a Developed India

- Captive Power plants in India have increased from 40 GWe in 2012 to almost 80 GWe in 2020 **(~2 times in 8 years).**
- These captive power plants are majorly utilised by energy intensive industries like Steel, Cement, Chemical, Metals and Minerals, Engineering, Textiles, Mining, etc.

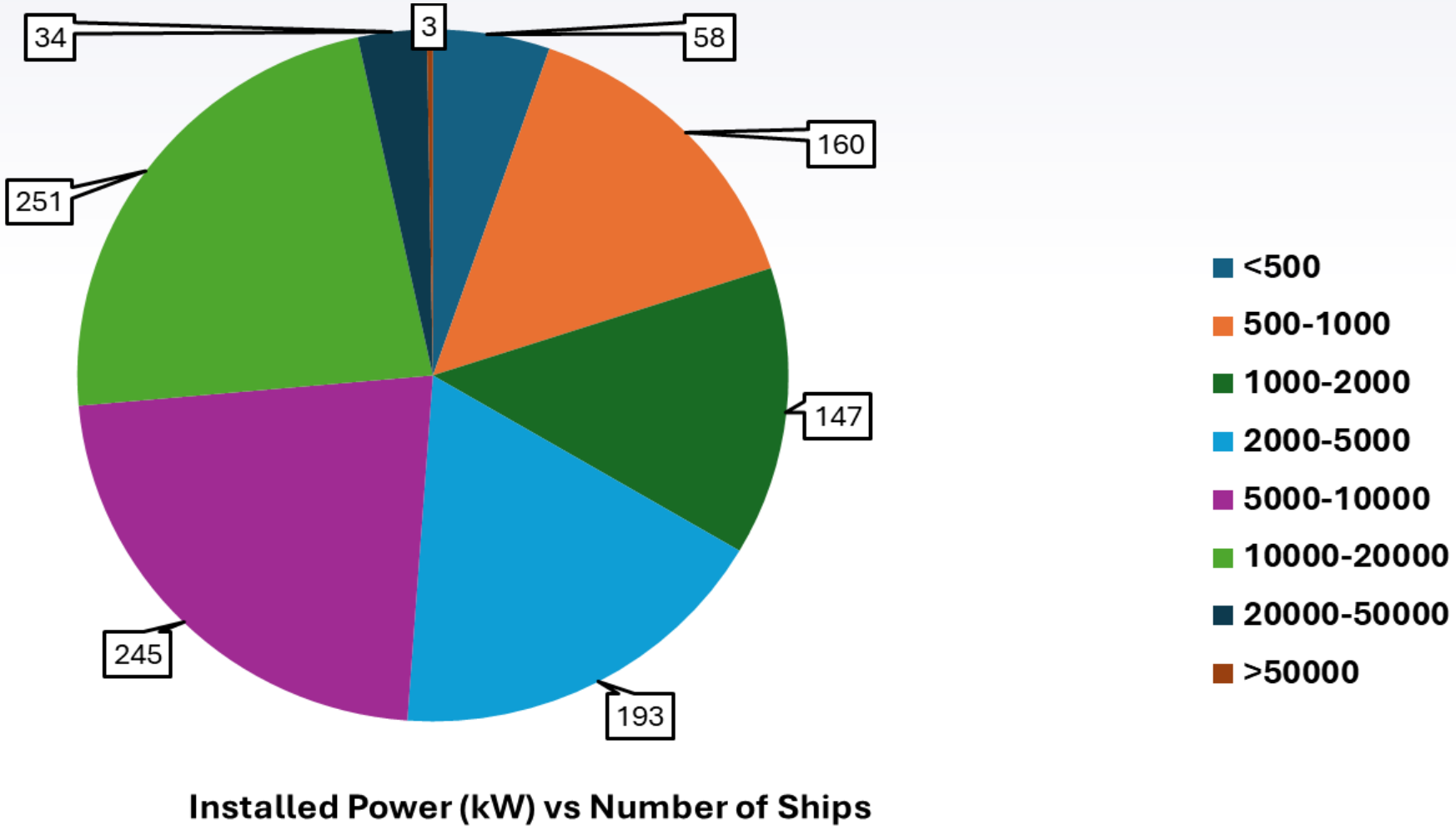


- It is envisaged that energy intense industries such as steel will grow by 8 times and cement by 3 times, if India to be developed. This will require at least 6-8 times more energy **(McKinsey, 2022)**.
- Thus, energy demand through captive power plants is estimated **to be approx. 400 to 500 GW by 2070**.

SMR & MMRs Demand in Transport Sector



SMRs & MMRs Demand in Cargo Shipping



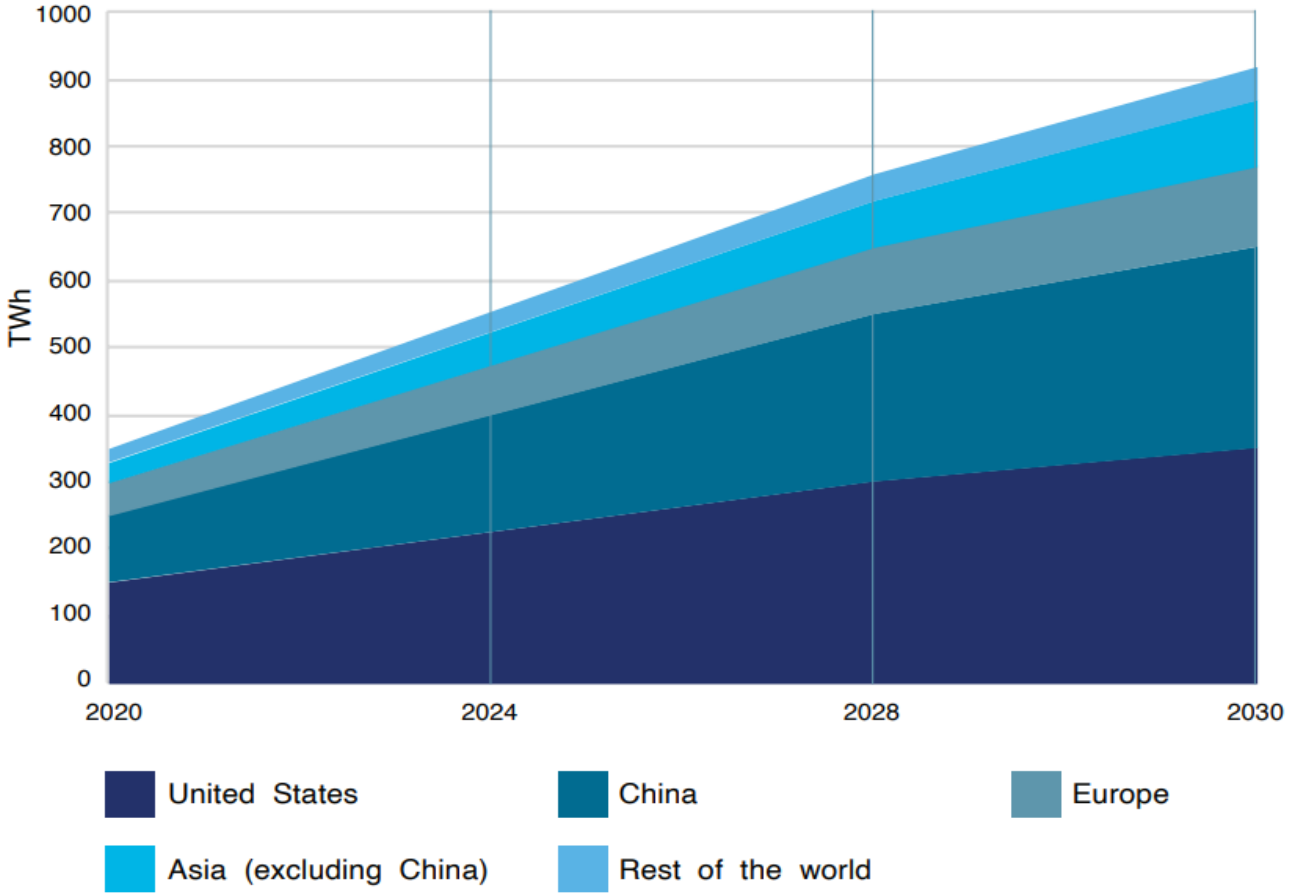
Installed Power (kW) vs Number of Ships

ORF OBSERVER RESEARCH FOUNDATION

The Consumption of Energy by Data Centres

Implications for the Global South

By 2047, India's data center capacity is projected to require approximately 35 GW of power to support rapid AI and cloud infrastructure growth, aligning with the "Viksit Bharat 2047" vision.



Source: International Energy Agency[28]

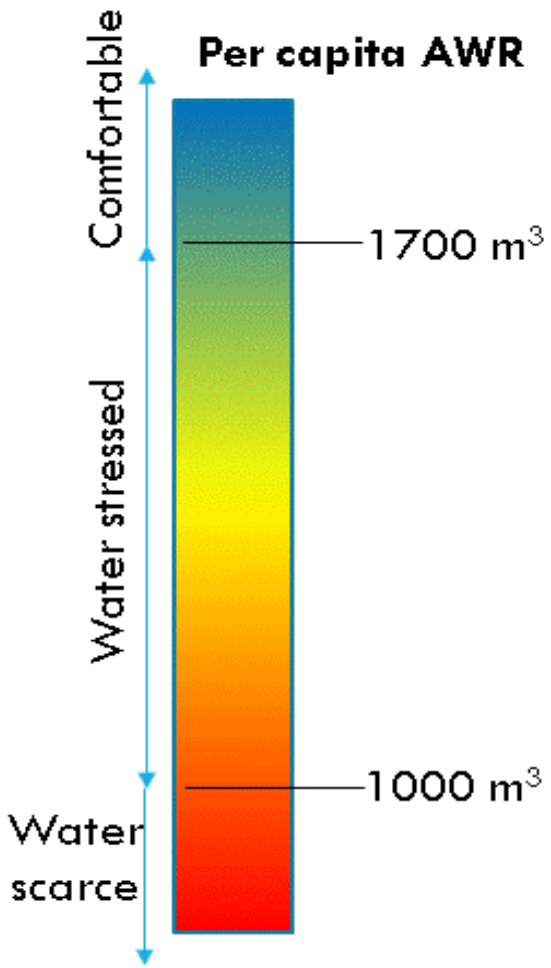
Other Energy Requirements

- Desalination for supplying clean drinking water:
  - India has 6100 km of coastline in the mainland itself. During low demand hours, the electricity from some of the units can be diverted for desalination of sea water using RO membrane.
  - Typically, per MWe, 6000 m^3/d of clean water can be produced by SMRs / MMR.

Per Capita Water Availability In India

Year	Population (Million)	Per capita water availability (m3/year)
1951	361	5177
1955	395	4732
1991	846	2209
2001	1027	1820
2025	1394	1341
2050	1640	1140

Source: Government of India, 2009.



Nuclear Power in India – where are we and what are targets?

Current Capacity and Production:

- India has 25 nuclear reactors in operation across 8 power plants, with a total capacity of 8,880 MW. Nuclear power generated 57 TWh of electricity.

Ongoing Expansion:

- Eleven more reactors are currently under construction with a combined capacity of 8,700 MW. This includes the PFBR and four 1000 MW Russian-designed reactors.

Future Goals:

- The government has set a target to achieve 100 GW of nuclear power capacity by 2047, This expansion will involve building more reactors and potentially introducing small modular reactors (SMRs). ***This is a brave and welcome step in the history of India; of course, country would require several hundred GW nuclear to decarbon and meet its energy requirements for 35 trillion economy***

N-Power - what are major issues?

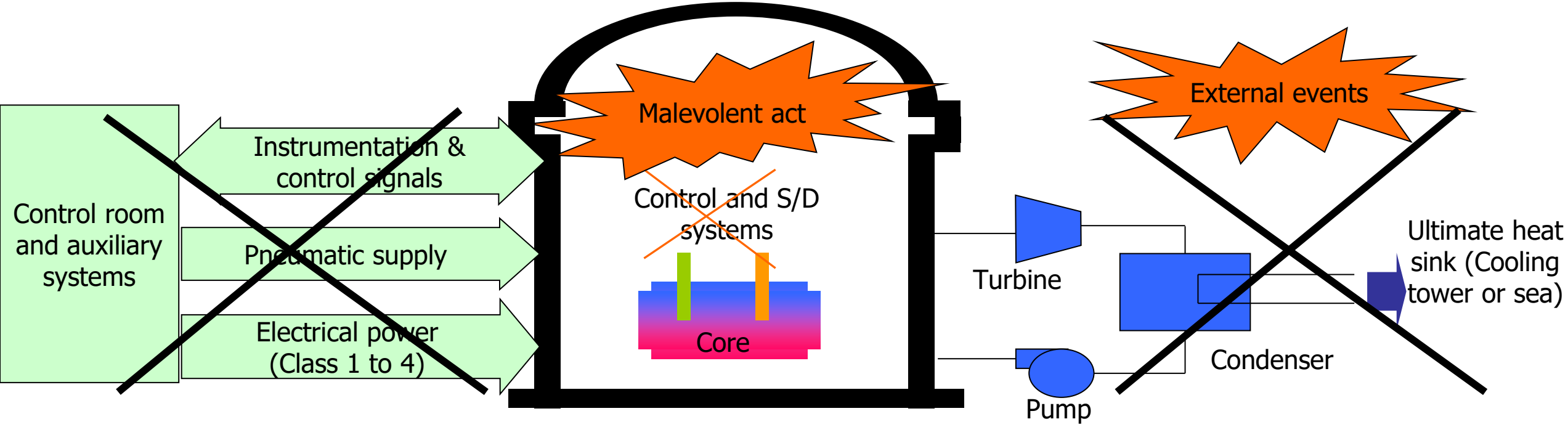
- **Cost – Capex and/or LCOE, unknowns**
- **Safety – Chernobyl & Fukushima - Radiation leak & Cancer**
- **Fuel supply security – Do we have enough uranium & when thorium to come**
- **Management of waste – each gram of irradiated uranium to be stored formore than 100,000 years - global concerns**

So a big debate, nuclear power - should we go or not to go - are we going towards "VINAS"

Lets us discover - Limitations of current commercially large nuclear reactors

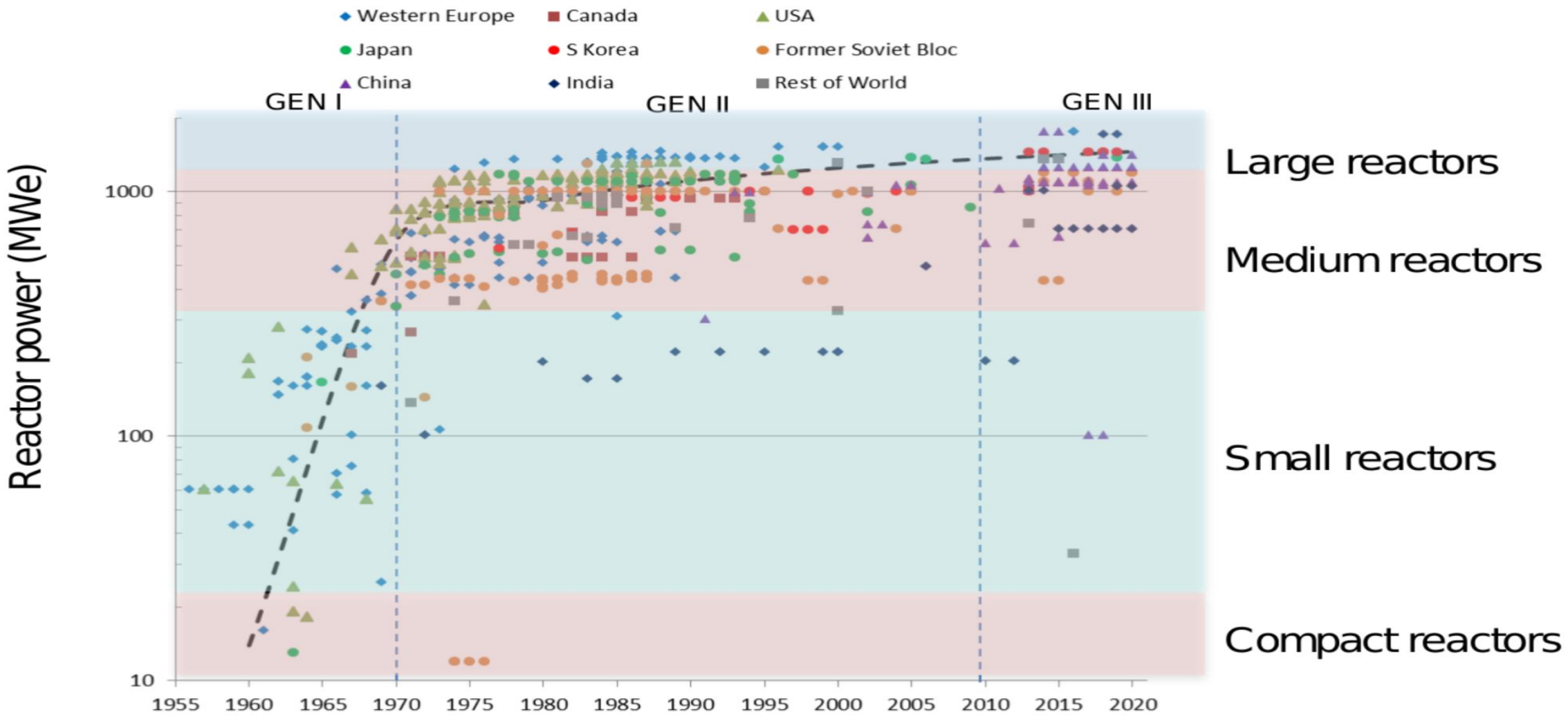
- **Acquiring several sites in India – qualifying seismic conditions - no seismic fault for 5 km radius – where to get???**
  - large population density of India
  - Requirement of ExZ (1 km rad), EPZ (16 km rad) to meet dose criteria of large plants
  - Coastal sites – availability is poor
  - Interior India – water availability, seismic zone qualification, seismic faults
- **Large project cost and construction delays**
  - Large investment because of large size of plants
  - Significant onsite work and civil construction delays
  - Non modularity in design, issues in delay in construction
  - High IDC (30 to 40%)
- **Safety issues of current reactors**
  - very low probable core melt accident may occur – Gen III plus designs have core catchers, hydrogen mitigation devices, containment venting – additional cost and safety issues remain – these reactors cannot satisfy no emergency planning in the public domain

Safety and Security – a higher level objective for large scale deployment scenario – limitations of current large conventional reactors



- No unacceptable radiological impact outside the plant boundary with
  - (a) Failure of all active systems, and
  - (b) Failure of external infrastructure to provide coolant, power and other services, and
  - (c) Malevolent acts by an insider, one of the consequences of which is the failure of instrumentation signal initiated shutdown actions, and
  - (d) Inability of plant operators to manage the events and their consequences, for a significantly long time.

Small Reactors Is not New - Water Cooled SMRs are Low Hanging Fruits

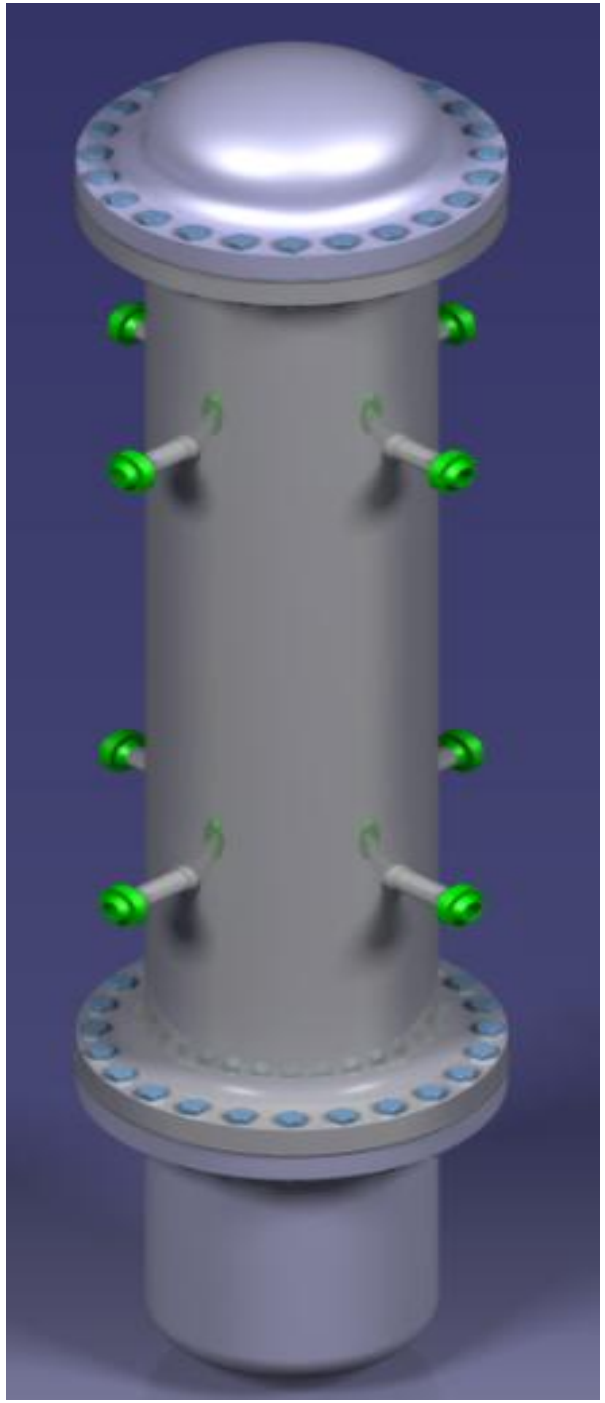


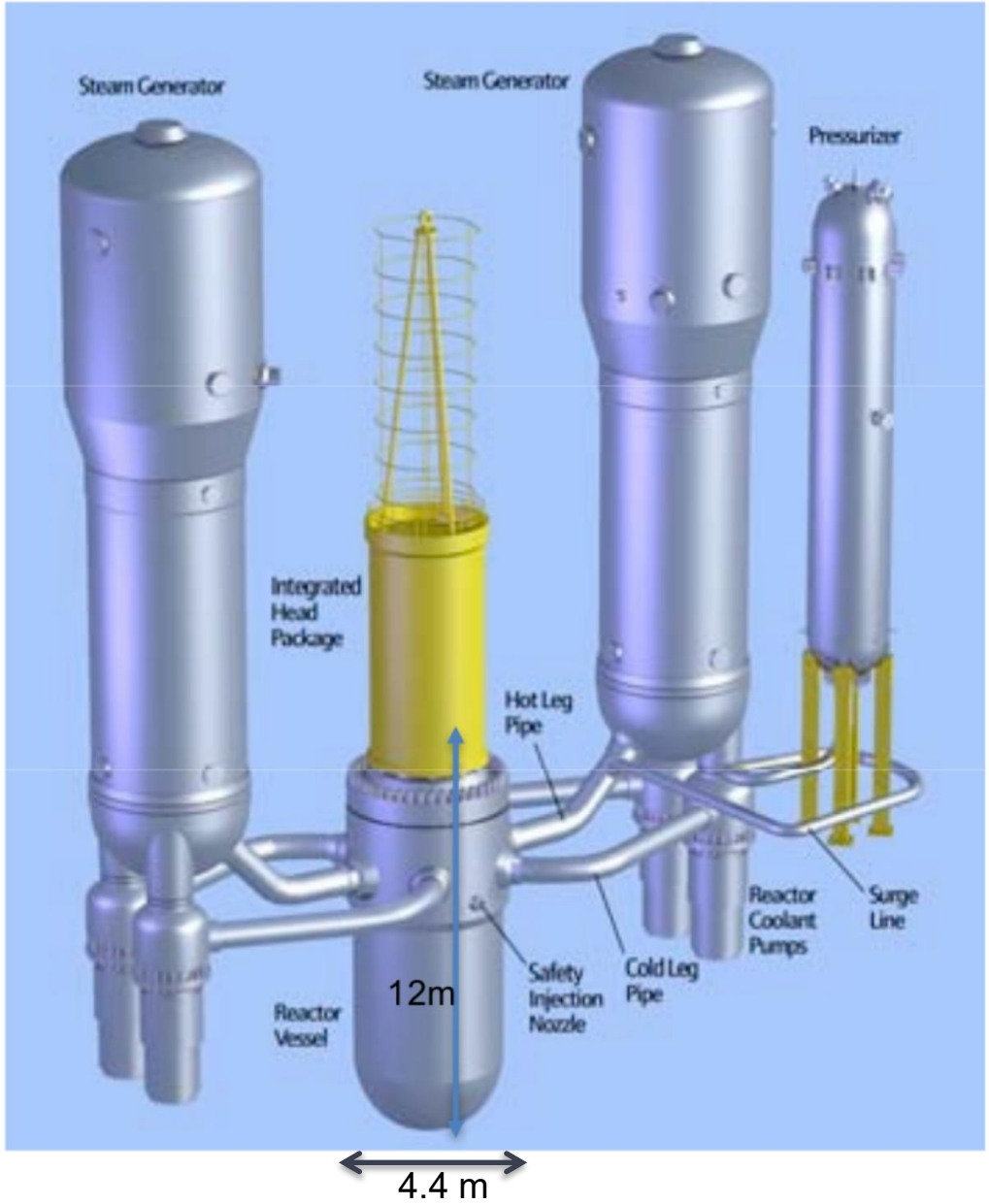
In the beginning there were only small reactors

What SMRs We look for?

Integral SMR – a single product from factory – very critical for accelerated deployment

- **Every reactor below 300 MWe is not an SMR**
- **Has to be resilient against internal or external sabotage**
- **Thorium use to be promoted – proliferation resistant**





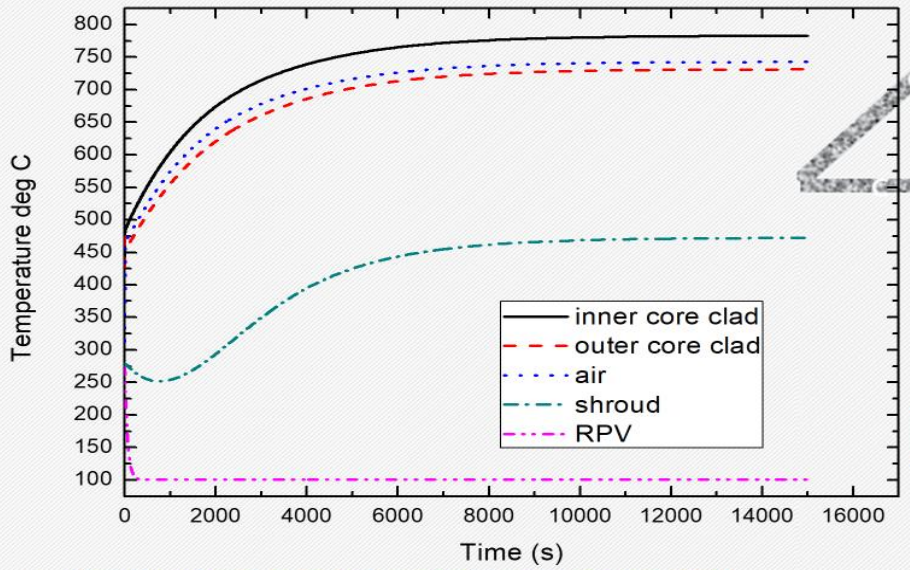
Conventional PWR

How SMRs Address the Issues of Current N-Technology

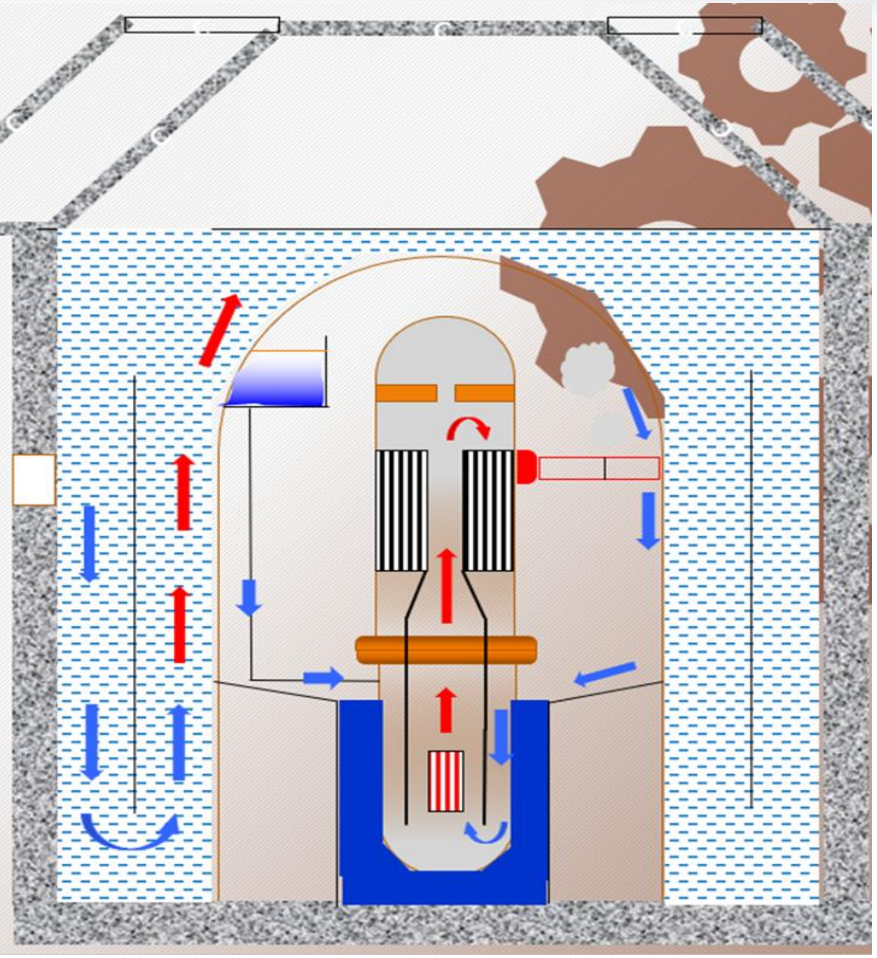
Enhanced Safety to accomplish no ExZ or EPZ requirement:

- These reactors carry almost zero risk to the public; They are designed with Trade Mark™ of “no impact in public domain”.
  - In SMRs, the size of reactor is small, radioactivity inventory is small, release at the plant boundary never exceeds allowable dose limit under any circumstances during the plant operation or even at shut down conditions requiring no emergency planning or evacuation in the public domain. i.e. ***“walk away safe reactor”.***

How much Safe is Safe Enough – a case study of a Passive Integral SMR?



- If all engineered safety systems fail, reactor still survives by natural circulation of air/steam within RPV.
- Maximum clad temperature is
  - 750 deg. C for 100 MWe,
  - 825 deg. C for 125 MWe and
  - 950 deg. C for 150 MWe plant.



SMRs for India - Whether Imported or Domestic

- Domestic manufacturing – 100% localization
- Possible to transport by rail or road – FOR INTERIOR INDIA
- Practically eliminate the core heat up – highly important to eliminate exclusion zone and emergency planning zone – Plant boundary is the ExZ for both Safety & Security.
- Minimum site work, mass production – target 36 months from FPC to Grid connection

SMRs for India - Whether Imported or Domestic

- Should be sited anywhere (high Seismic design basis)
- Dry cooling
- Flexibility to use Thorium fuel
- Complying to Closed fuel cycle

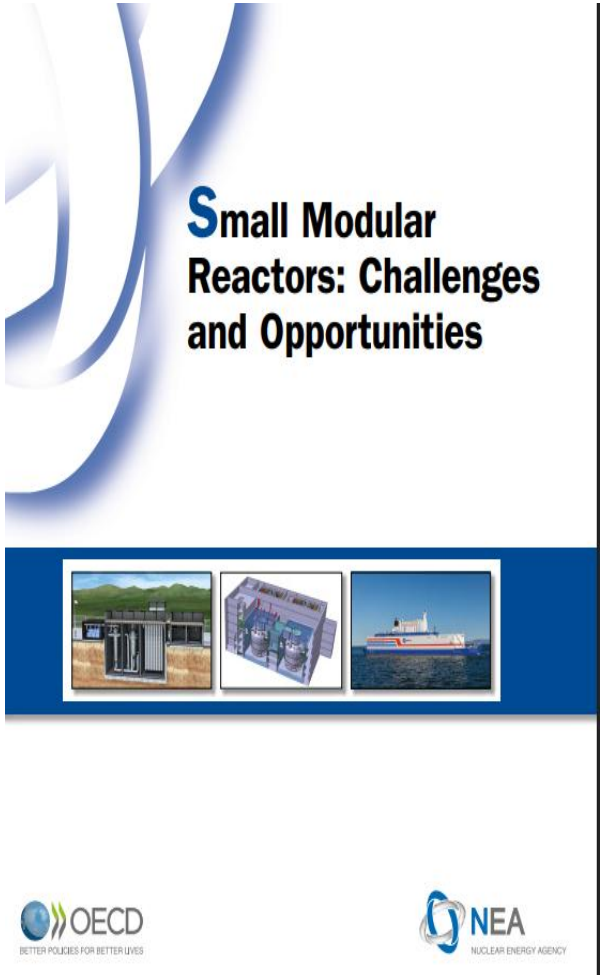
SMRs third party liability – May be excluded

5.2. Nuclear third party liability and SMRs

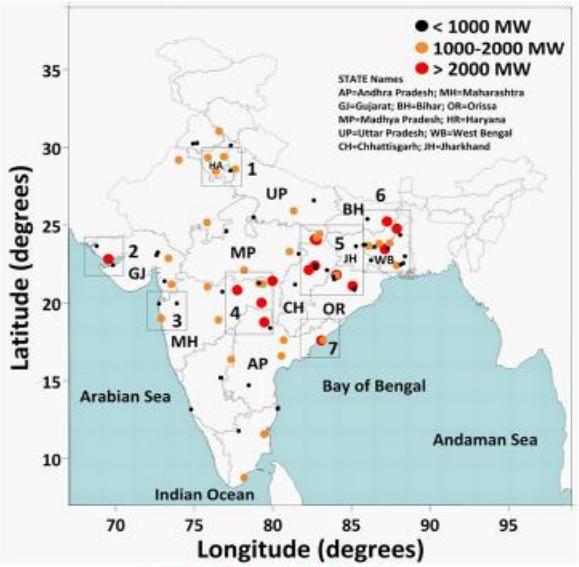
The international conventions governing nuclear third party liability are:

- the Paris Convention on Third Party Liability in the Field of Nuclear Energy ("Paris Convention" or PC),[11] which will soon be amended by a 2004 Protocol yet to enter into force ("Revised Paris Convention" or RPC);[12]
- the Vienna Convention on Civil Liability for Nuclear Damage ("Vienna Convention" or VC);[13]
- the Vienna Convention as amended by the 1997 Protocol ("Revised Vienna Convention" or RVC);[14]
- the Convention on Supplementary Compensation for Nuclear Damage (CSC).[15]

SMRs are included in the definition of "nuclear installation" provided in the conventions, which covers "reactors other than those comprised in any means of transport".[16] Having regard to the nature of the nuclear installation involved and to the likely consequences of a nuclear incident originating therefrom, the conventions (except for the Vienna Convention) allow countries to establish a lower amount of liability for that installation, provided that in no event shall any amount so established be less than the amounts provided in the conventions for low-risk installations.[17] The aim of this option is to avoid burdening the nuclear operators concerned with unjustified insurance or financial security costs.[18] Therefore, SMRs may be considered as low-risk installations if the installation states' applicable convention(s) and national laws allow for such a case.



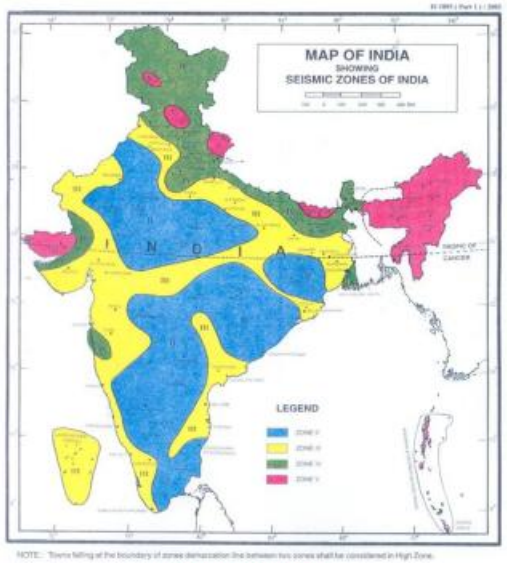
Potential of SMRs for Replacing Coal Plants for Electricity Production



COAL PLANTS LOCATIONS

India Coal Fired Power Plants 2013

http://www.urbanemissions.info/india-emissions-inventory/emissions-in-india-coal-fired-power-plants/

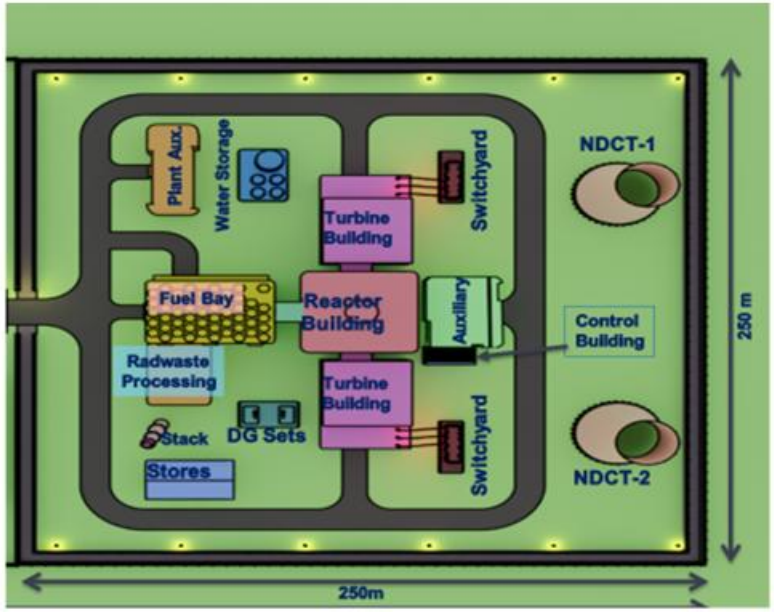


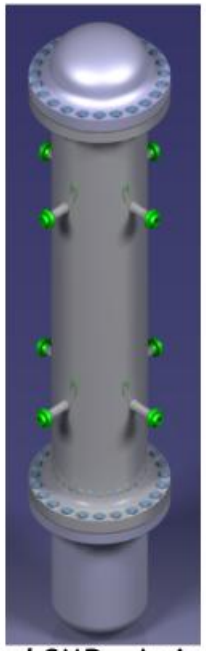
SEISMIC ZONES

MINISTRY OF EARTH SCIENCES
LOK SABHA UNSTARRED QUESTION No. 2046

Conclusion of this map comparison: Majority of the coal power plants are situated in zone II; and some are in Zone III. Hence, they can be easily replaced by nuclear plants.

Footprint of an SMR





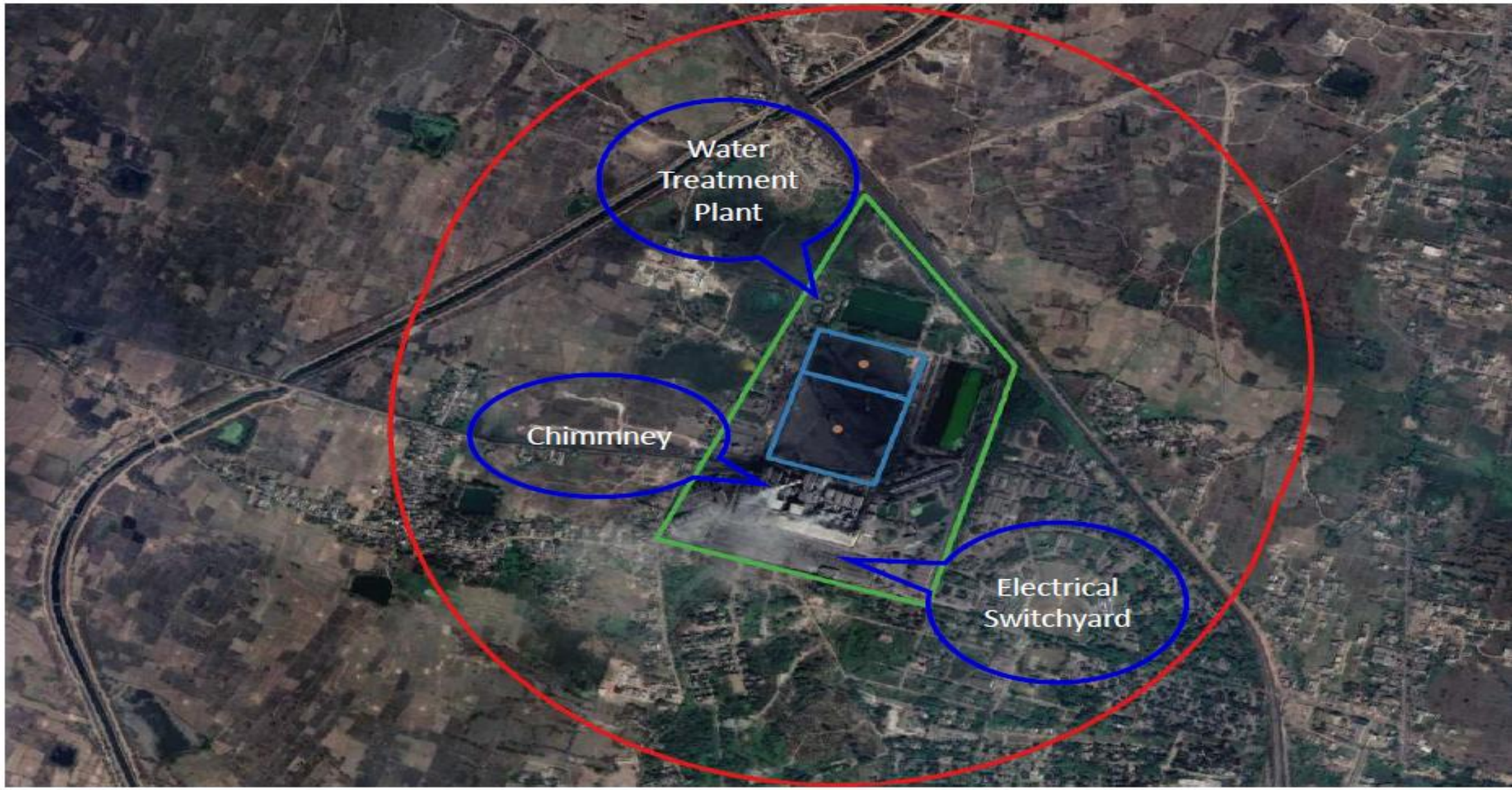
Integral SMR - A single product from factory

Recent NIAS-BARC Study under BRNS

Talcher TPS

Existing TPP
4x60+2x110
= **460 MW**

Proposed SMR
4x100+2x100
= **600 MW**

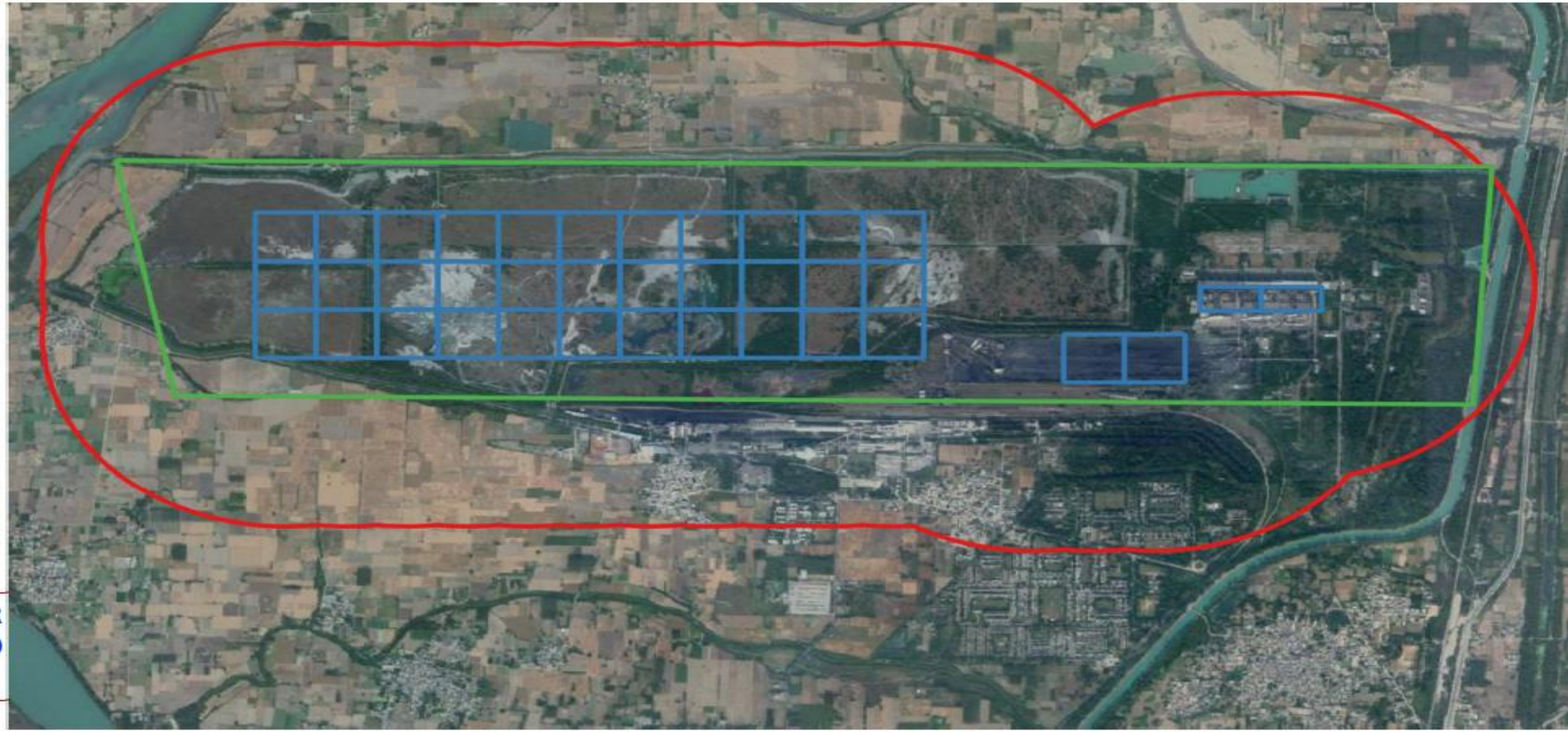


NIAS-BARC Study under BRNS

Guru Gobind Ropar TPS

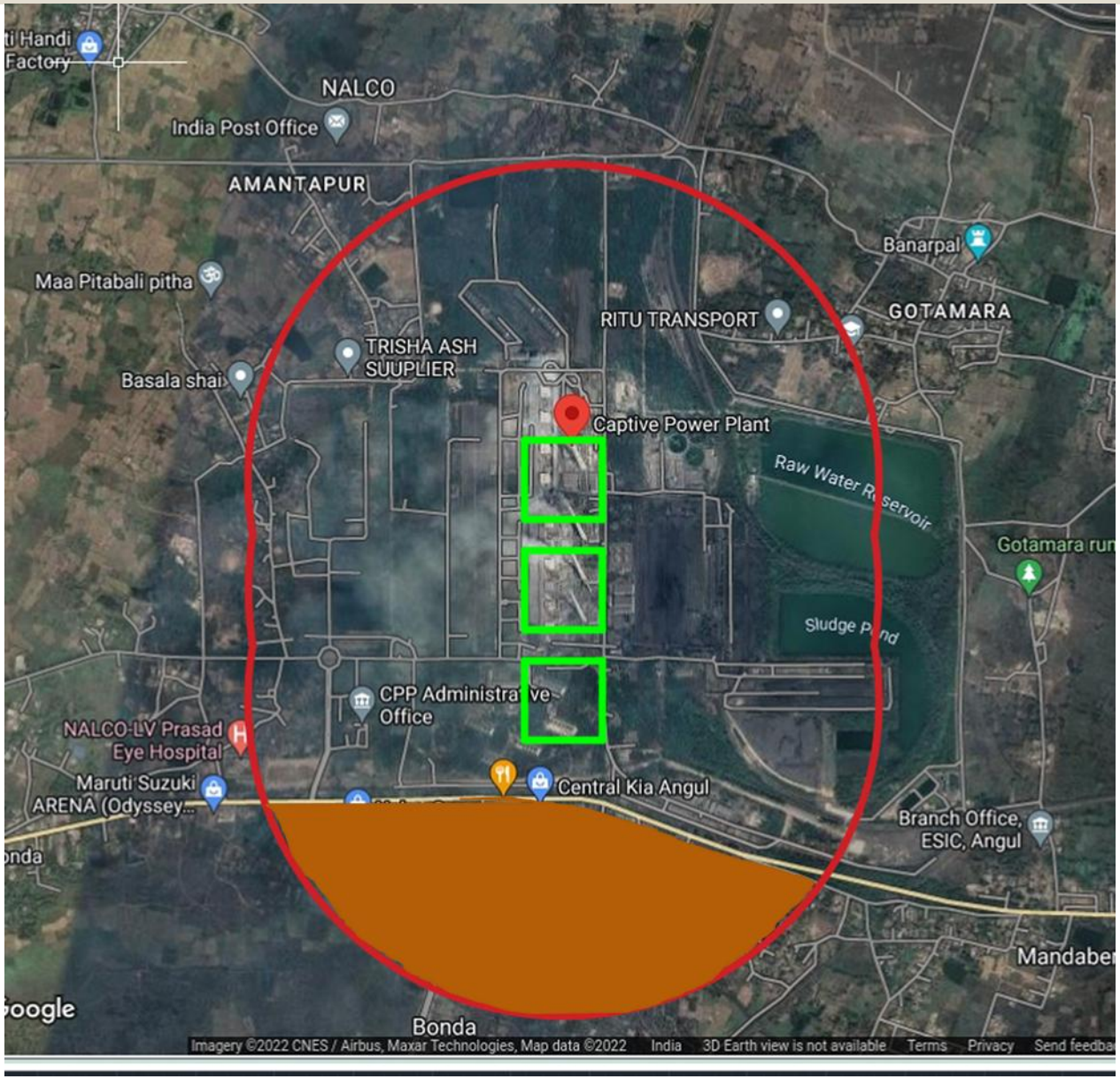
Existing TPP
6x210
= **1260 MW**

Proposed SMR
36x400+2x200
= **14,800 MW**



NALCO Captive Power plant 10 X 120 MWe

Possibility can be increased to
3 X 400 MWe or 6 X 400 MWe



Super imposing SMR modules at Uppur Site

32 modules

Capacity will be 32x400 = 12800 M

(12.8 GW)

SMR Park

Saving Factors by Choosing Existing Coal Sites for Nuclear Power

Sr. No.	Activity	Savings		Remarks
		Time	Cost	
1	Land acquisition for Plant	2-5 Years	Nil	Already at utility possession
2	Water Source	2-3 Years	Nil	Already at utility possession with higher capacity than SMR requirement
3	Site survey	2-5 Years	Nil	DPR of Plant already exist only revalidation required
4	Transport facility (Rail and Road connectivity)	1-2 Years	5 % of site activity	Already Exists Only security needs to improvise
5	De-commissioning of Boiler and turbine		NIL	
6	Trained Manpower	--	Scrap value of plant is a asset	Except nuclear operation the BOP operation already available
7	Township for Employees and local facilities	1-3 Years	--	Majority exists only capacity enhancement and modernization required
8	Sewage treatment plant	2-5 Years	10 % of Project cost	Already available

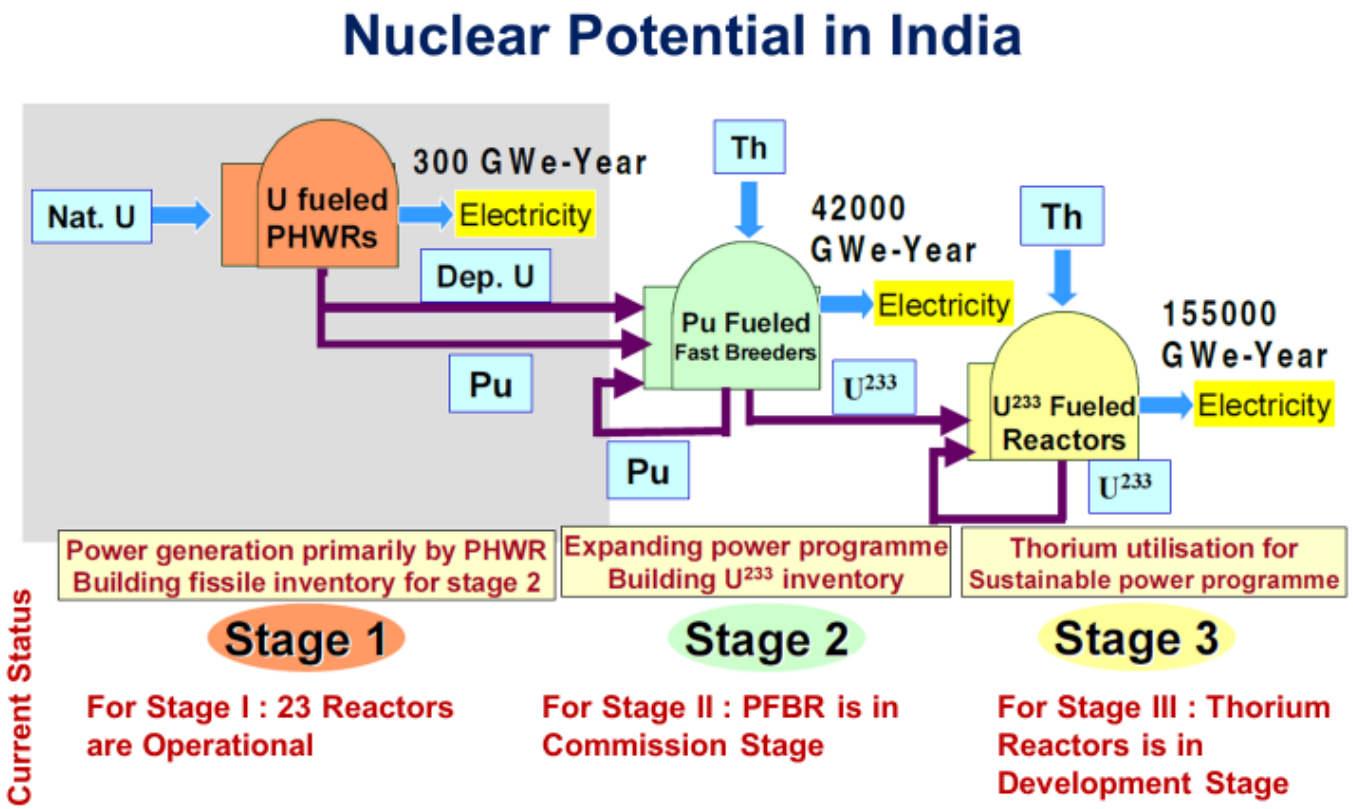
*The timeline is indicative and taken from open literatures and engineering thumb rules

Potential of Micro Modular Reactors for Hydrogen Production and Rapid Deployment in NE Region, Data Centres, Maritime industries, Isolated regions, Malls, Hospitals, Airports, etc. for Uninterrupted Power Supply

- Power less than 20 MWe
- Drive Cargo Ships – Several thousands
- Data Centres – Compact Uninterrupted Power Supply
- Small Captive Power Plants require clean/ green energy for production of Hydrogen/ localized electricity supplu
- High temperature MMRs– ideal for Hydrogen production - for Transport and Industry energy consumption:
  - Typically, per MWe, around 480 kg of hydrogen can be generated per day using standard low temperature electrolysis. If high temperature steam electrolysis is used, 630 kg of hydrogen can be generated per day per MWe by dedicated SMRs/Micro Reactors.

Nuclear Potential in India through Domestic Fuel Reserves

Fuel



- The first stage targets to generate close to 10 GW electricity through NU.
- The second stage envisages generating close to 42000 GWe-years (~4,00,000 TWh) through Pu FBR.
- The final stage is the development of advanced nuclear power reactors targeting energy production of 155000 GWe-years (~14,00,000 TWh) through thorium fuel.
- If we consider the need for 100 years, the second and third stage combined can provide nearly 18000 TWh/year.
- Further can be enhanced through imported fuel.

Accelerating Thorium Deployment

- **PHWRs (BSR & 700 MWe)**
- **AHWRs (LEU-Th) (has potential to be converted to SMR)**
- **ADS**

Establish the thorium front end & back end – job by DAE institutions

Deploy Advanced SMRs/MMRs using thorium in different platforms :

- **MSBRs**
- **GCRs**
- **Other technologies (Hybrid)**

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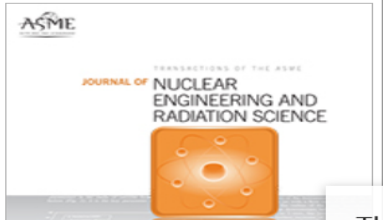


RESEARCH-ARTICLE

Pressurized Heavy Water Reactor Technology: Its Relevance Today

A. K. Nayak, S. Banerjee

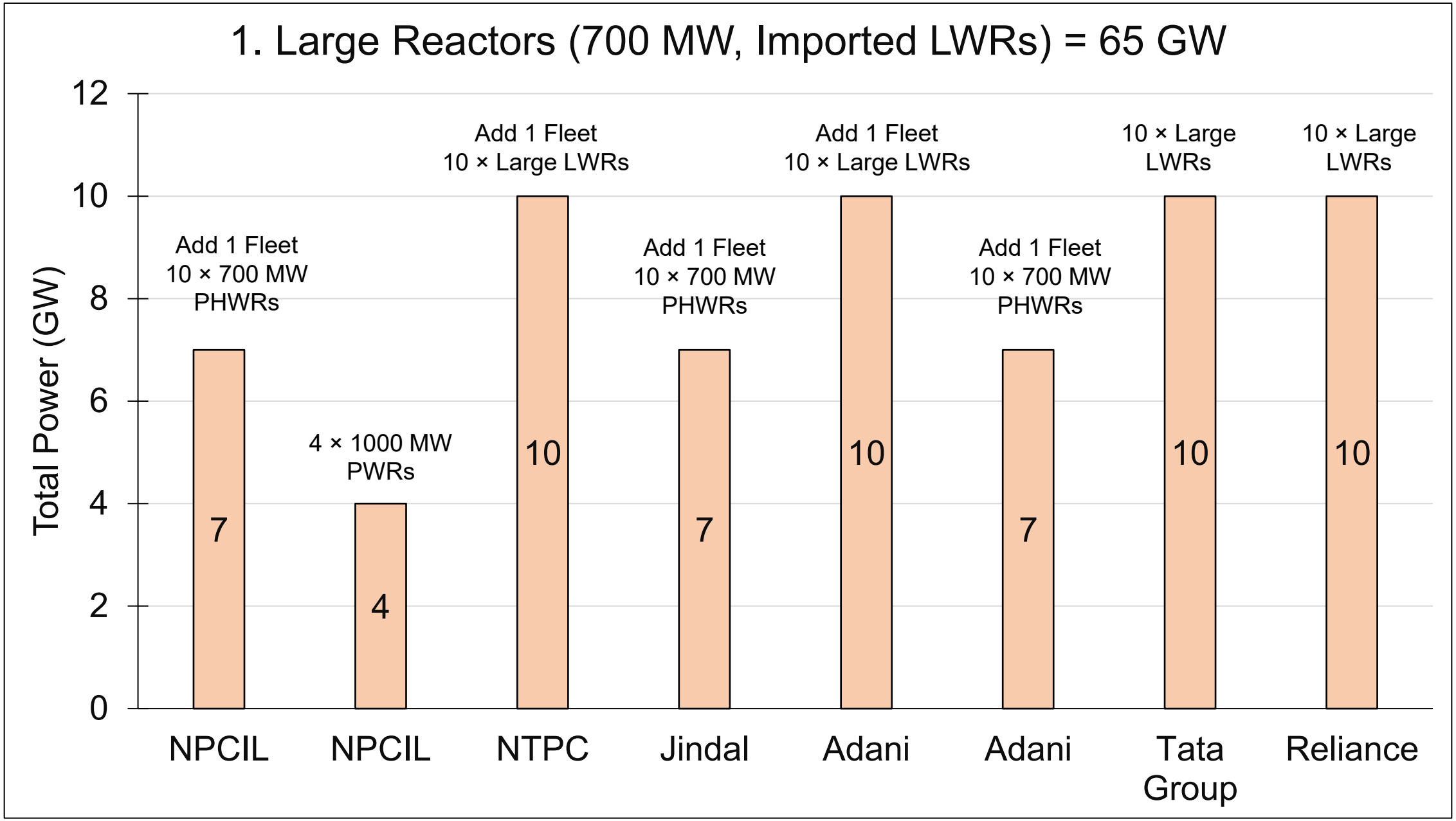
Check for updates

+ Author and Article Information

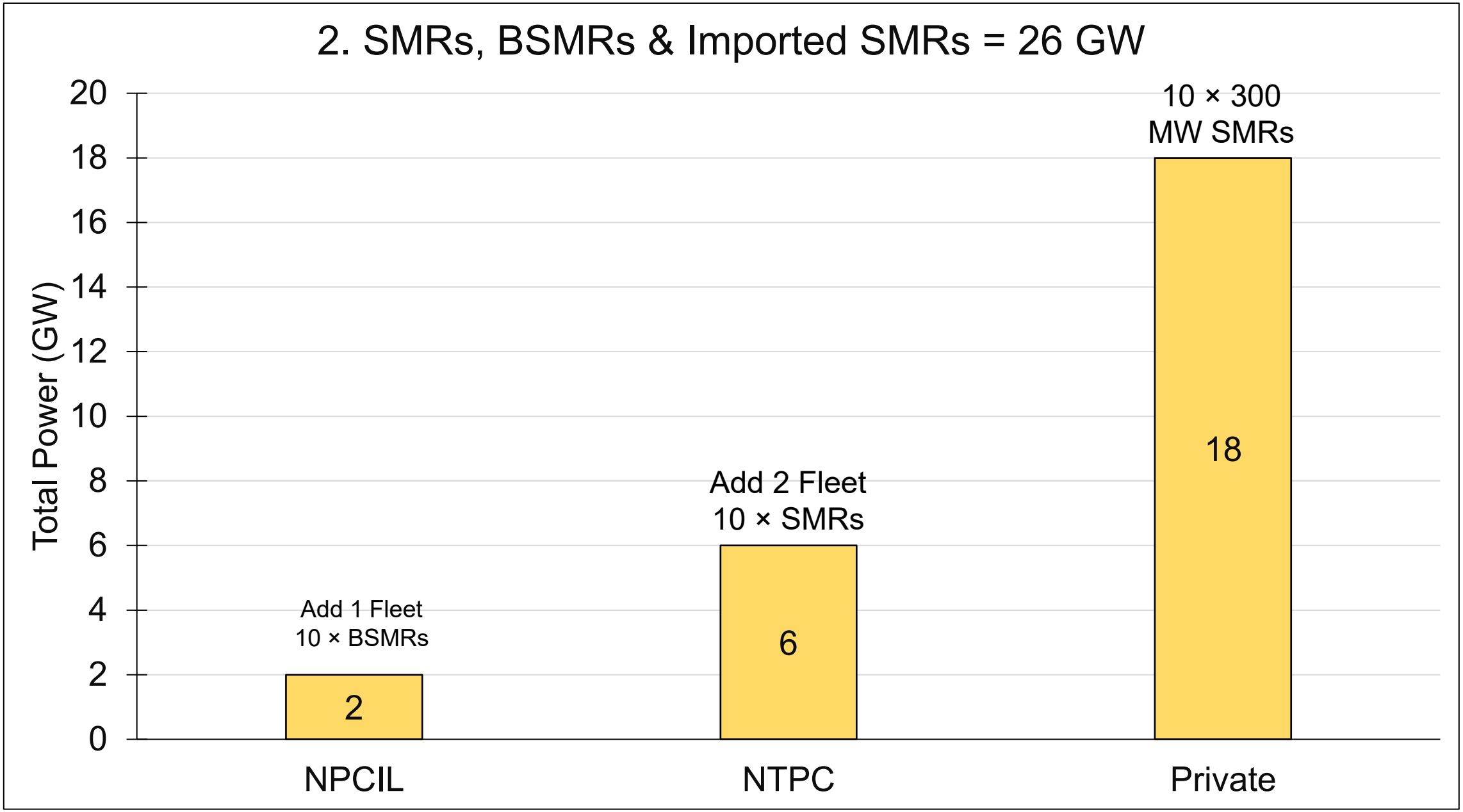
In a CANDU reactor, the single-pass, Th-LEU fuel cycle increases uranium utilization by up to 30% relative to natural uranium fuel, and would provide a low-cost insurance policy against future shortages of uranium as well as provide a bridge to the closed thorium cycles of the future.

Wilfrid Bennett Lewis - recognized as the **"Father of the CANDU reactor"**

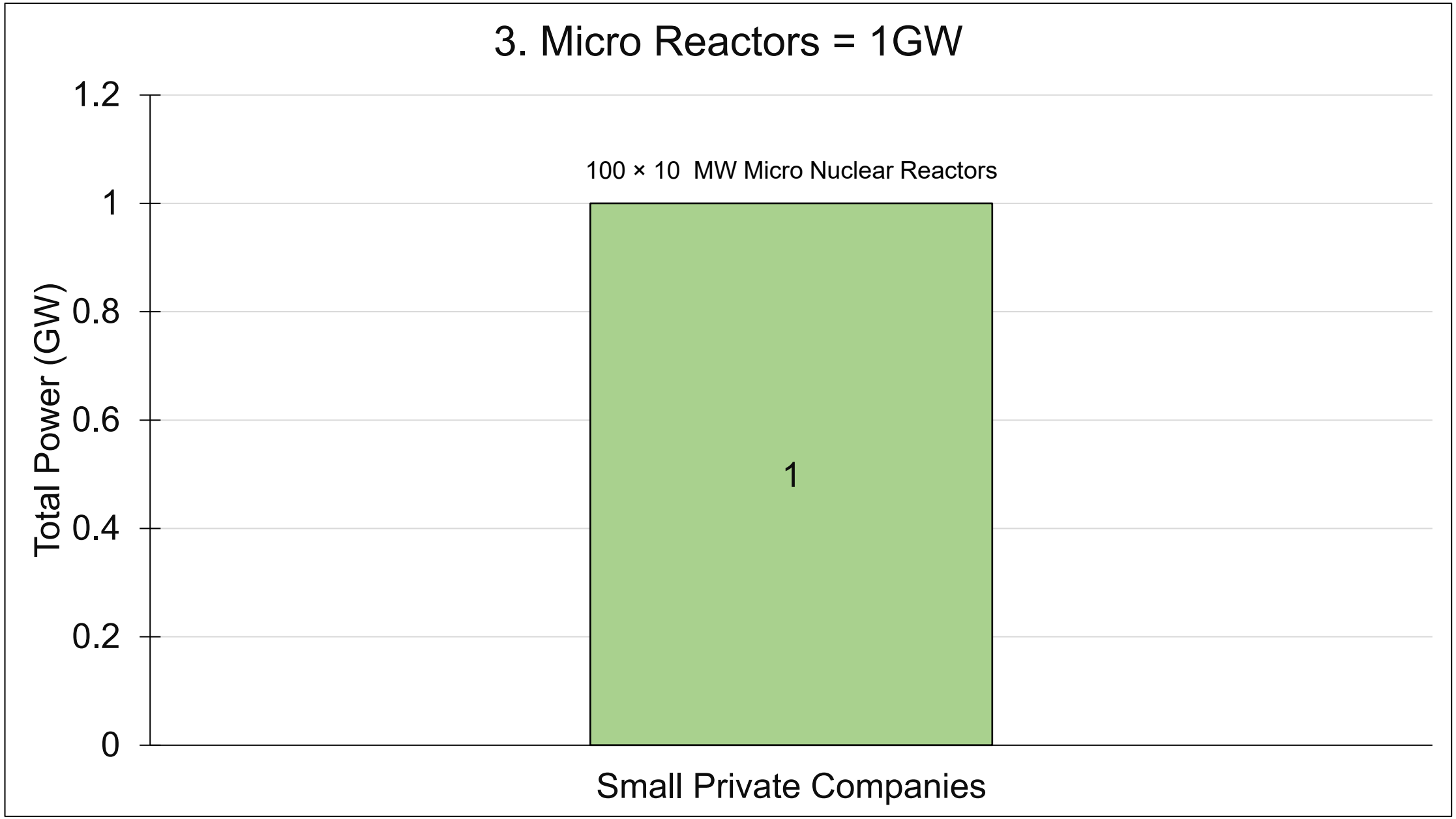
Strategic Road Map for 100 GW



Strategic Road Map for 100 GW



Strategic Road Map for 100 GW



Task Ahead (1) - Technologies & Vendors Establishment

Task Ahead (2) - Fuel Supply

Task Ahead (3) – Establishment of fuel fabrication facilities

Task Ahead (4) - Supply Chain Development

Task Ahead (5) - Financing

Task Ahead (6)- Licensing aspects

Task Ahead (7) - Human Resources Development & Capacity Building

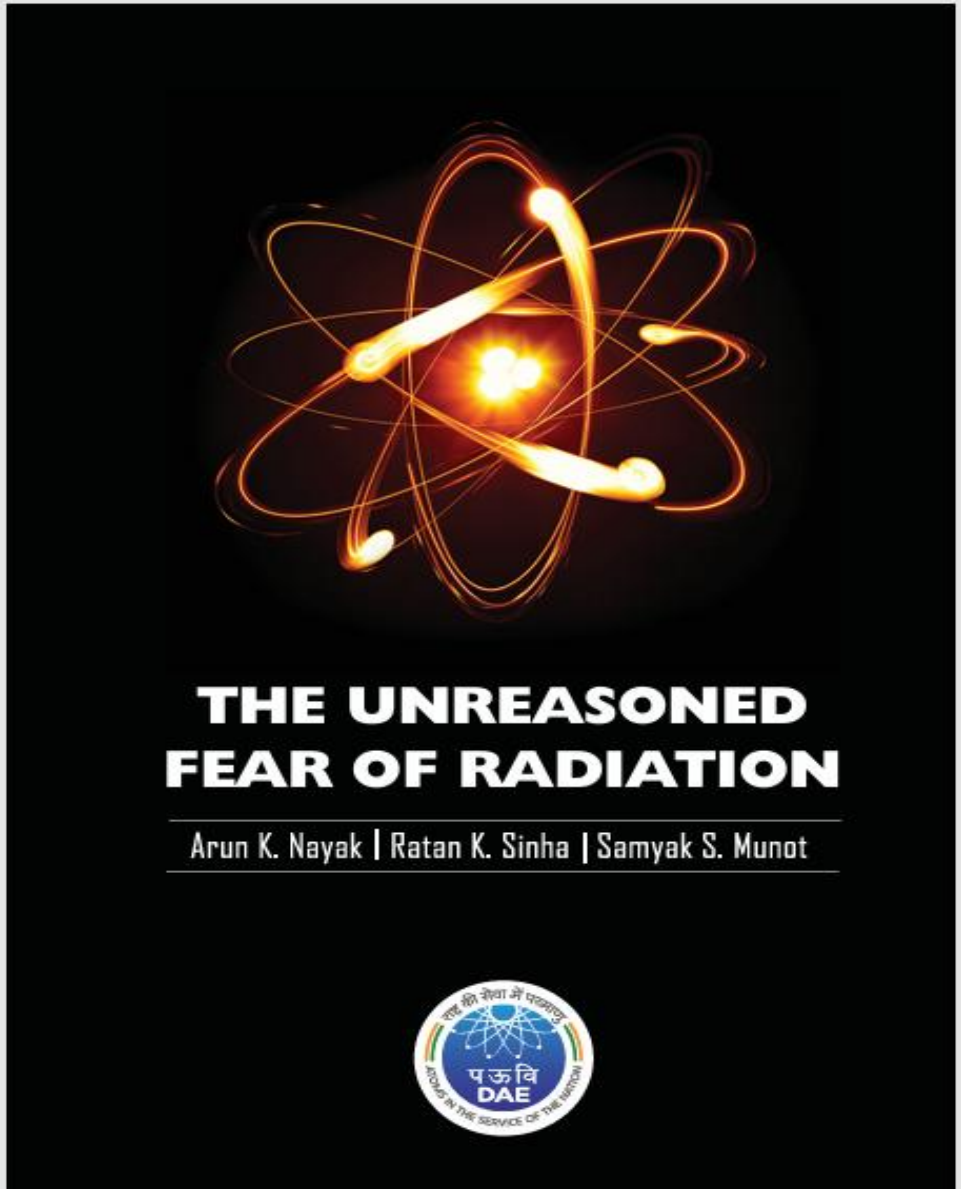
How SHANTI Act will help for accelerated deployment

- Private, Central/State PSUs allowed to build, own, operate including decommission NPPs – a dramatic change in nuclear power expansion
- Private funding including foreign investment to accelerate n-power growth
- any industry can import/fabricate the fuel – burden on DAE & NFC removed
- Collaboration with foreign entities to deploy reactors of Gen III+, Gen IV, SMRs and MMRs in accelerated manner
- Safety, Security & Safeguards remain in Government Control – so public concerns eliminated
- Licensing remain in Government Control – only those licensed to enter into n-power business – public concerns eliminated

How SHANTI Act will help for accelerated deployment

- Rad waste to be handled by operator under AERB safety REGULATION – public concerns eliminated
- AERB now established by ACT – concerns of independent functioning of AERB eliminated
- Civilian Nuclear energy research & development opened up – many start-ups, collaborations expected soon
- Third party liability simplified by removing ambiguities in 17(b) & 46 of CLND Act; Liability amounts layered according to the size of nuclear power plants & facilities

Revisiting Dose Criteria



Key Aspects of the ALARA Decision:

- **The Change:** The DOE is removing ALARA from directives, including DOE Order 458.1 and 10 CFR 835, which will alter long-standing safety-first protocols.
- **Reasons for Removal:** The policy was viewed by the administration as overly burdensome to the nuclear industry with minimal safety benefits, aiming to align regulations with current, less conservative scientific evidence.

Thank you

R. K. Laxman's famous cartoon depicting Pandit Nehru driving the common man on a bullock cart with a nuclear wheel – on the eve of the dedication of the first nuclear research reactor APSARA (1957)